



Weather Data Analysis - Final Project Report

Distributed Processing Course

Location: Cairo, Egypt (30.0444°N, 31.2357°E)

Analysis Period: 2000 - 2024 (25 years)

Date: December 2024

1. Executive Summary

This report presents a comprehensive analysis of 25 years of weather data from Cairo, Egypt, demonstrating distributed processing concepts through an ETL pipeline and Map-Reduce implementation. The analysis reveals a clear **warming trend of +1.36°C per decade** with 2024 being the hottest year on record at 46.4°C.

Key Findings

Metric	Value
Total Observations	9,132 daily records
Hottest Year	2024 (46.4°C)
Coollest Year	2006 (40.6°C)
Mean Annual Max Temperature	43.24°C
Temperature Trend	+1.36°C per decade
Anomalies Detected	1 (2024)

2. Project Objectives

The project focused on:

1. **Gathering raw weather data** (temperature, precipitation, wind speed)
 2. **Cleaning and preparing the dataset** using ETL principles
 3. **Computing statistical and analytical features** from raw values
 4. **Applying Map-Reduce** to compute yearly maximum temperatures
 5. **Presenting extracted features** in clear, visual forms
 6. **Detecting anomalies** in weather patterns
-

3. Methodology

3.1 Data Pipeline Architecture

EXTRACT → TRANSFORM → LOAD → MAP-REDUCE → ANALYSIS → VISUALIZATION

1. **Extract:** Downloaded data from Open-Meteo Archive API
2. **Transform:** Converted JSON to structured CSV format
3. **Load:** Stored processed data for analysis
4. **Map-Reduce:** Computed yearly maximum temperatures
5. **Analysis:** Applied statistical methods for insights
6. **Visualization:** Generated 12 professional charts

3.2 Map-Reduce Implementation

The Map-Reduce pattern was implemented to find the **highest temperature of each year**:

- **Map Phase:** Extracted (year, temperature) pairs from each daily record
 - **Shuffle Phase:** Grouped temperatures by year
 - **Reduce Phase:** Computed maximum temperature for each year
-

4. Analysis Results

4.1 Comprehensive Dashboard

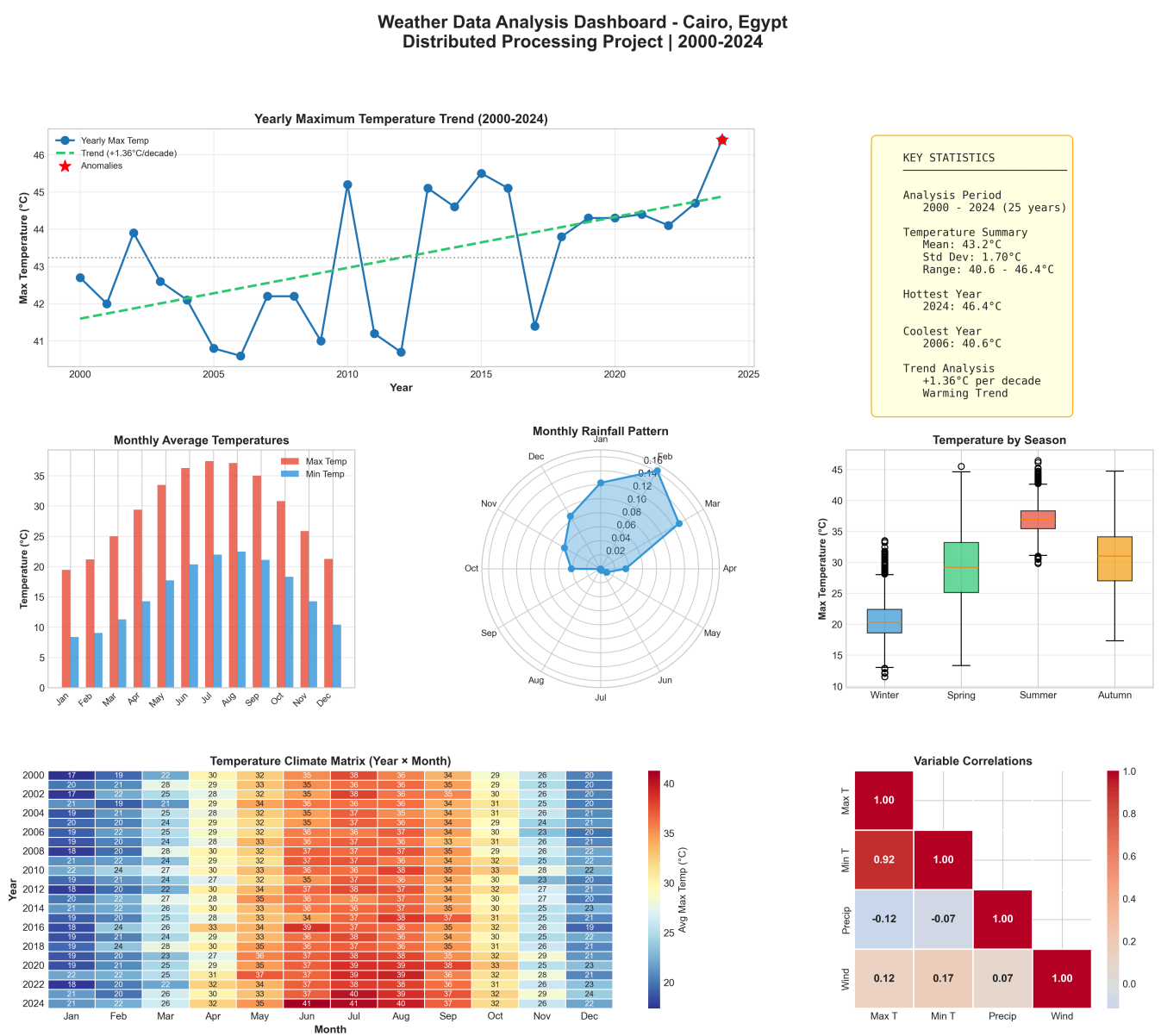


Figure 1: Comprehensive analysis dashboard showing key metrics, trends, and patterns

4.2 Yearly Temperature Trend Analysis

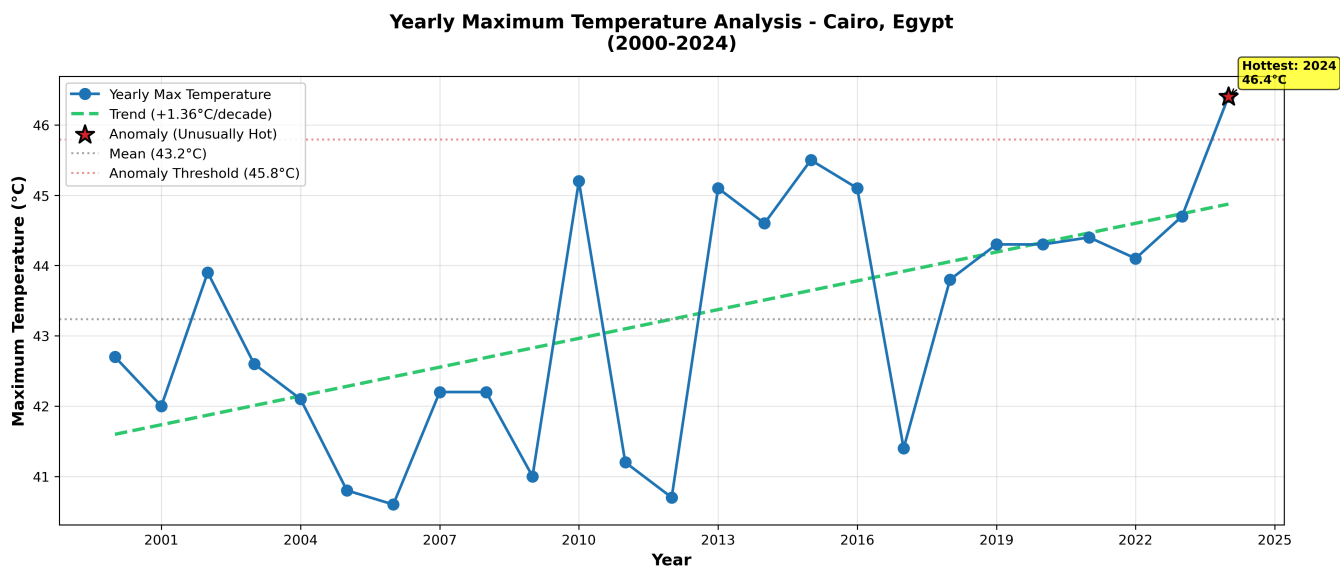


Figure 2: 25-year temperature trend with linear regression and anomaly detection

Key Observations:

- Clear warming trend: **+1.36°C per decade**
- Temperature range: 40.6°C to 46.4°C
- 2024 identified as an anomaly (above 1.5 standard deviations)

4.3 Temperature Distribution

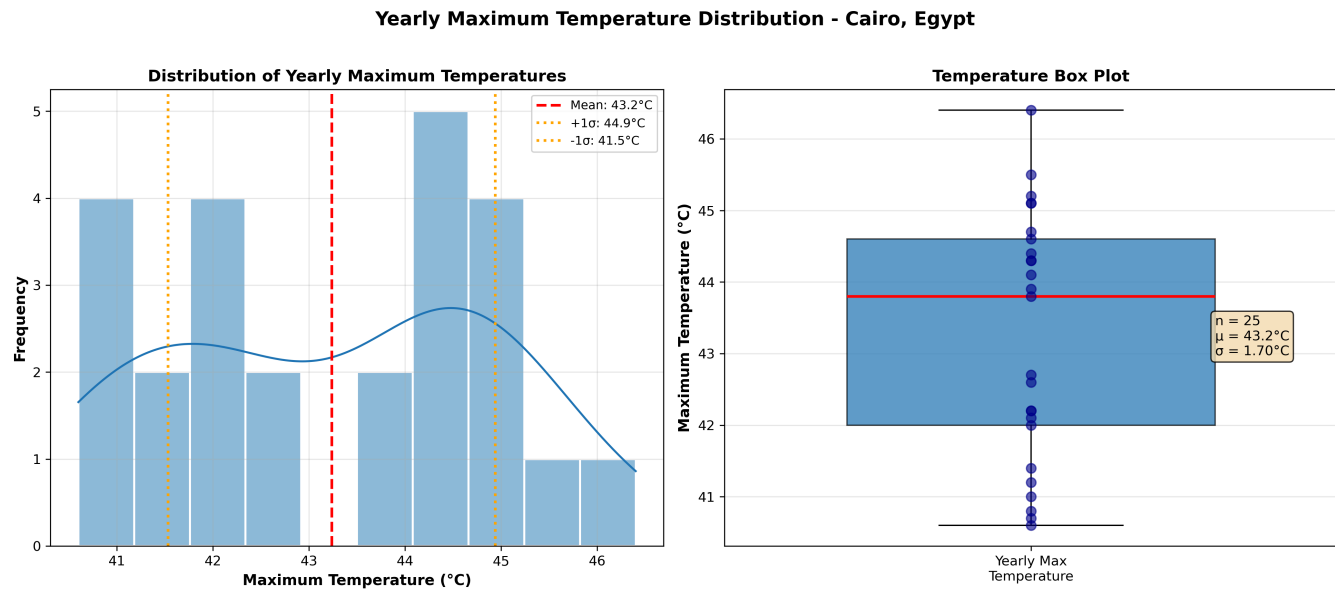


Figure 3: Distribution of yearly maximum temperatures with statistical markers

Statistical Summary:

- Mean: 43.24°C
- Standard Deviation: 1.70°C
- Distribution shows right-skew indicating increasing temperatures

4.4 Decade Comparison

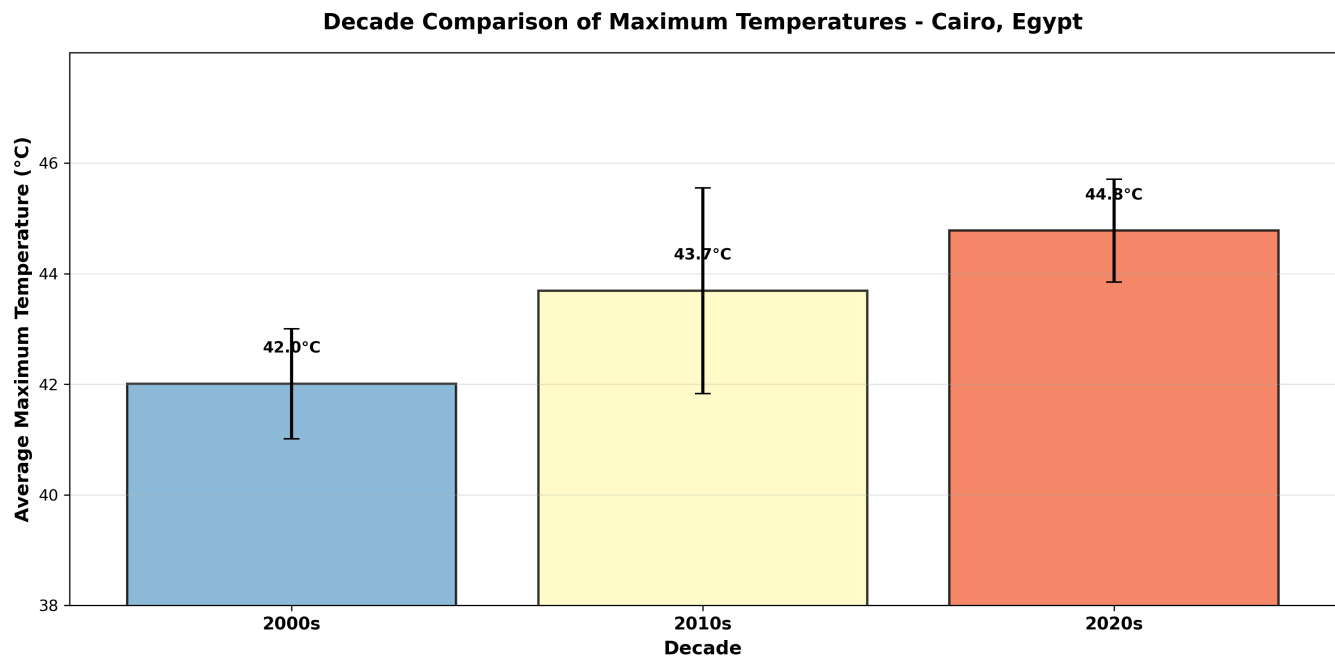


Figure 4: Comparison of average maximum temperatures by decade

Decade Progression:

- 2000s: Cooler baseline period
 - 2010s: Significant warming
 - 2020s: Continued warming trend with record temperatures
-

4.5 Monthly Temperature Patterns

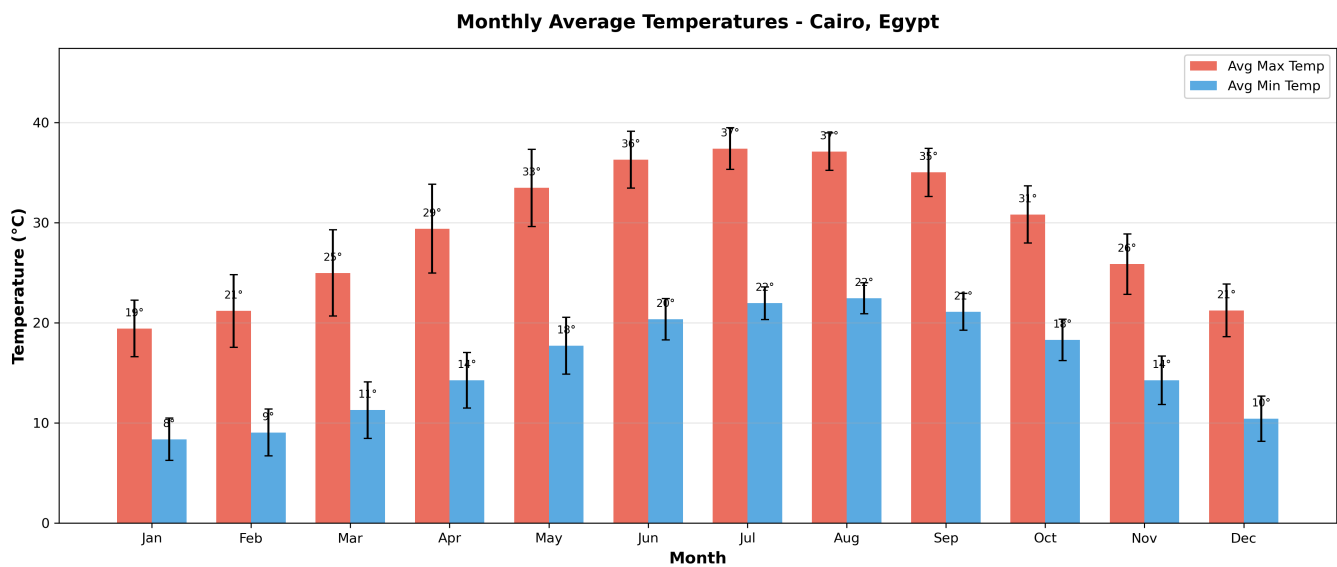


Figure 5: Monthly average temperatures showing seasonal variation

Monthly Extremes:

- Hottest Month: July (37.4°C average max)
- Coldest Month: January (19.4°C average max)
- Temperature Range: 18°C between summer and winter

4.6 Annual Temperature Cycle

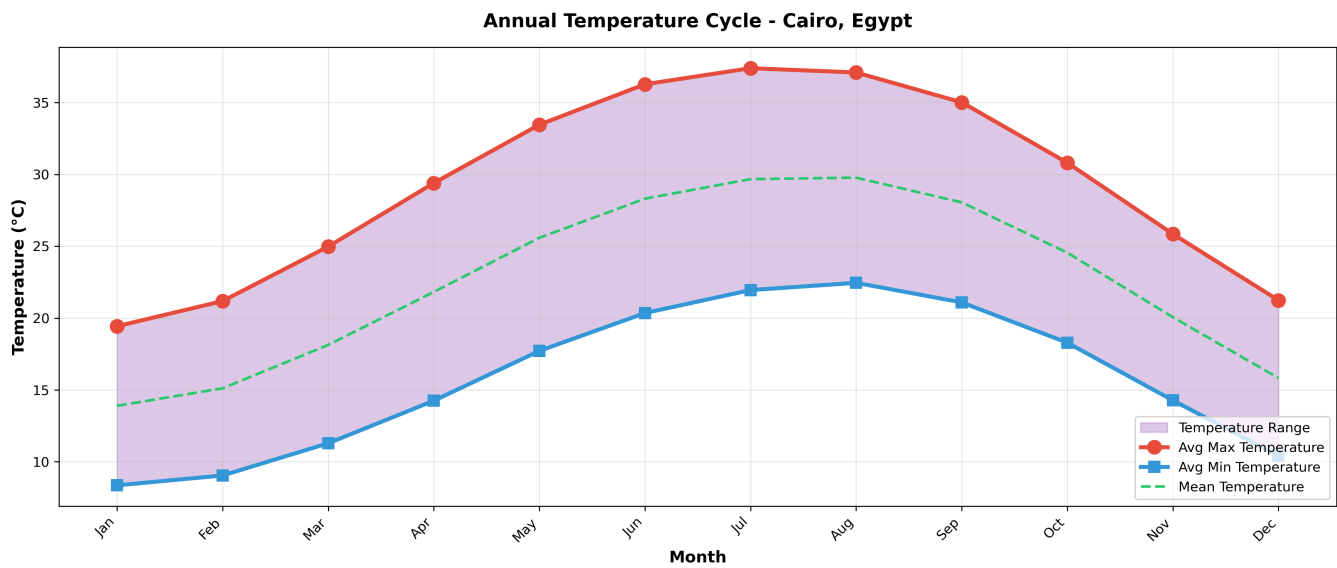


Figure 6: Annual temperature cycle showing the transition between seasons

4.7 Rainfall Distribution (Polar Chart)

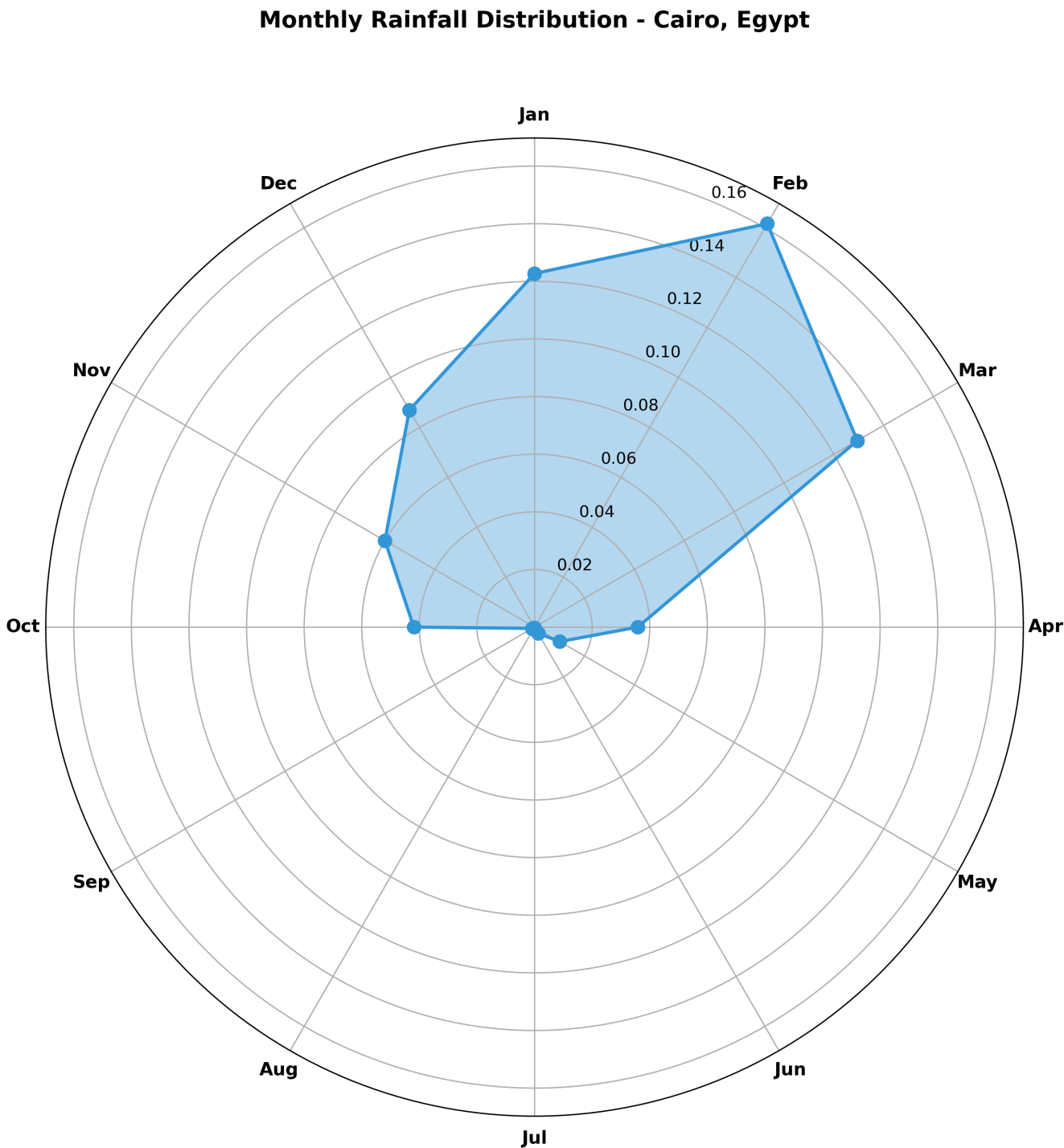


Figure 7: Monthly rainfall distribution in polar format

Precipitation Pattern:

- Wettest Month: February (0.2mm/day average)
- Driest Month: July (nearly 0mm)
- Cairo exhibits typical Mediterranean/desert climate with winter rainfall

4.8 Monthly Precipitation

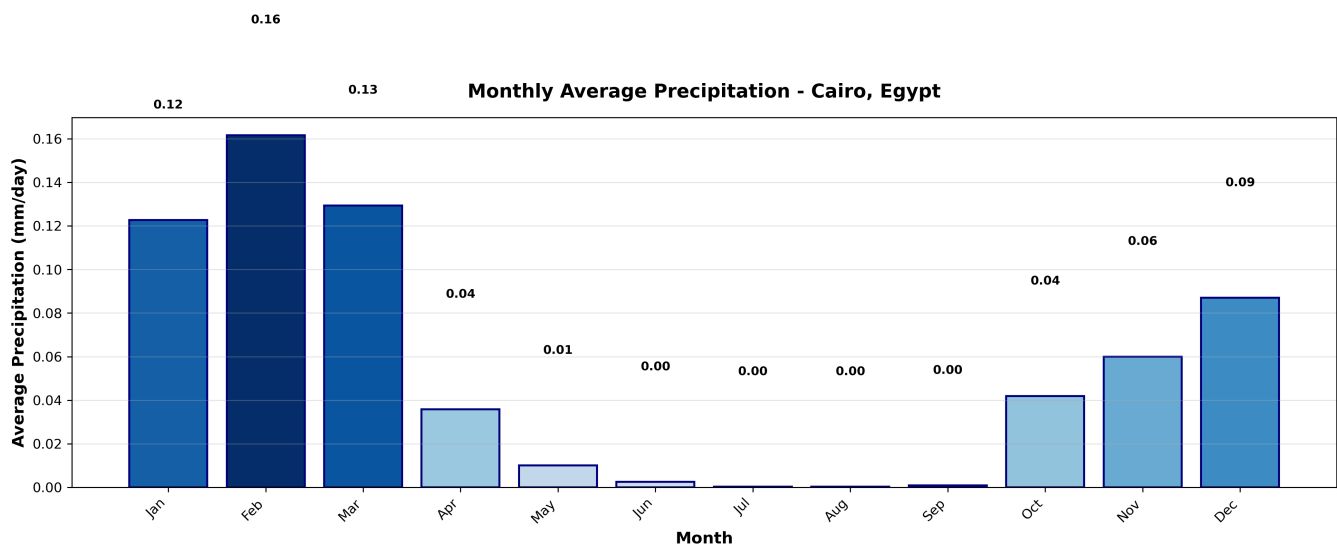


Figure 8: Bar chart of monthly precipitation averages

4.9 Wind Speed Climate Heatmap

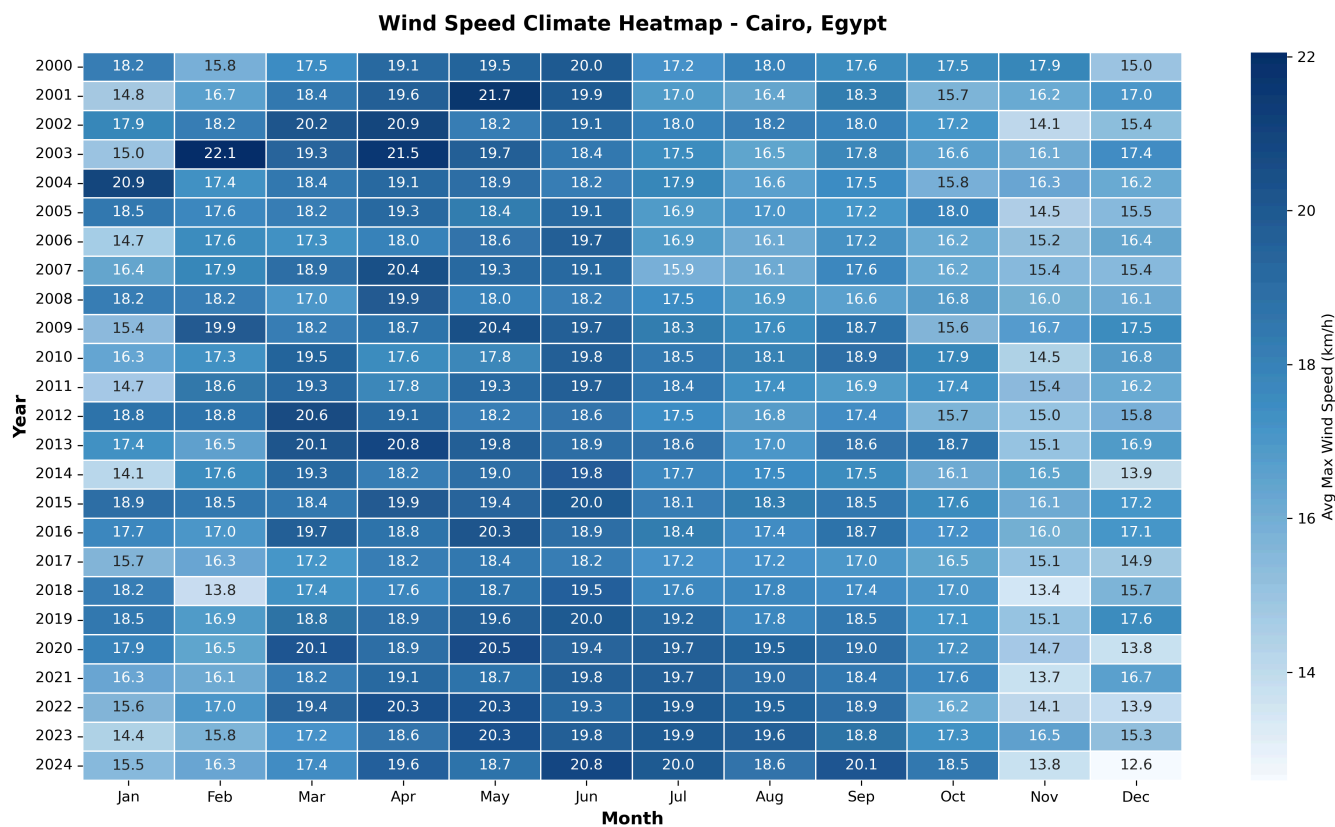


Figure 9: Year × Month heatmap of average maximum wind speeds

Wind Patterns:

- Windiest Month: June (19.4 km/h average)
- Higher wind speeds during spring/summer transitions
- Consistent patterns across years

4.10 Temperature Climate Heatmap

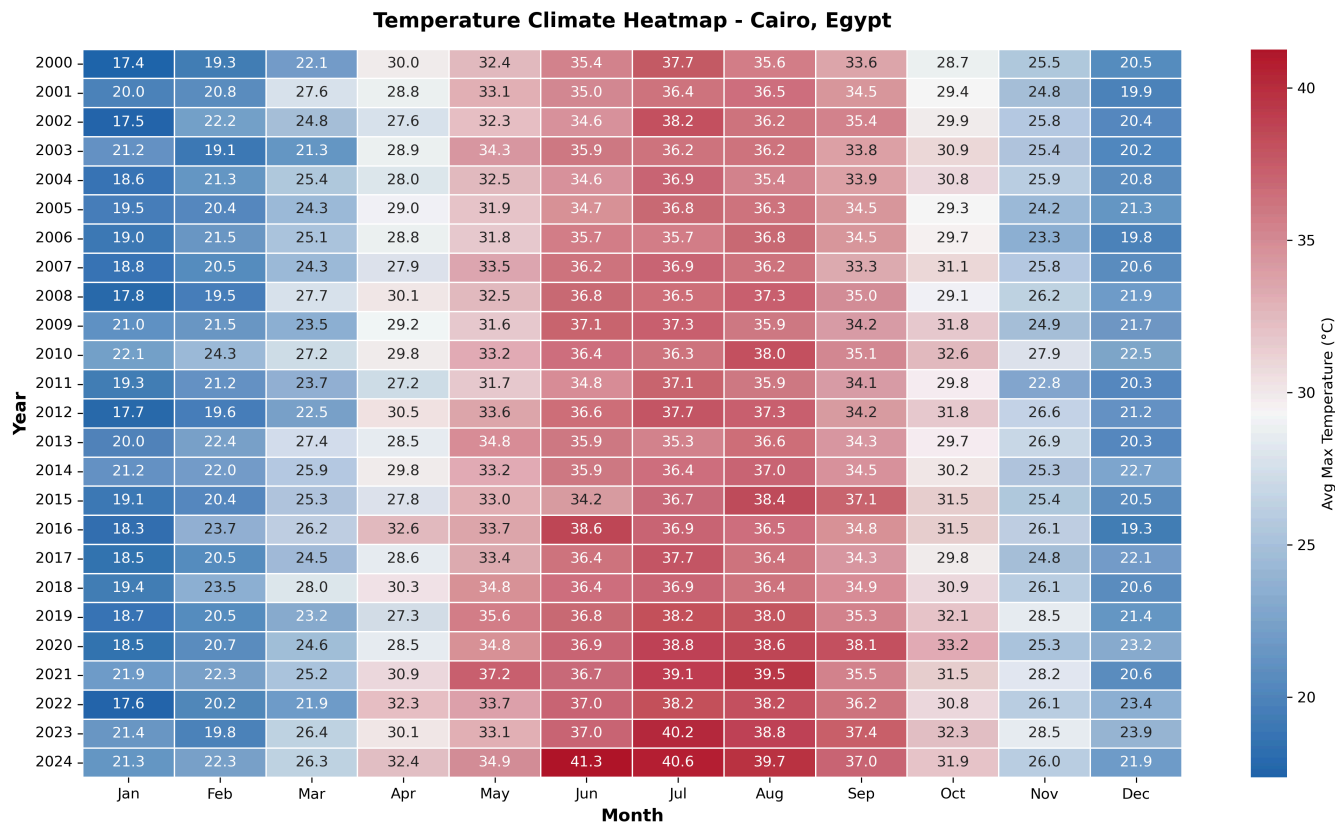


Figure 10: Year × Month heatmap showing temperature patterns over 25 years

Visual Insights:

- Clear seasonal bands (cooler winter months, hotter summer months)
- Gradual intensification of red colors in recent years (warming)
- July-August consistently shows the highest temperatures

4.11 Seasonal Comparison

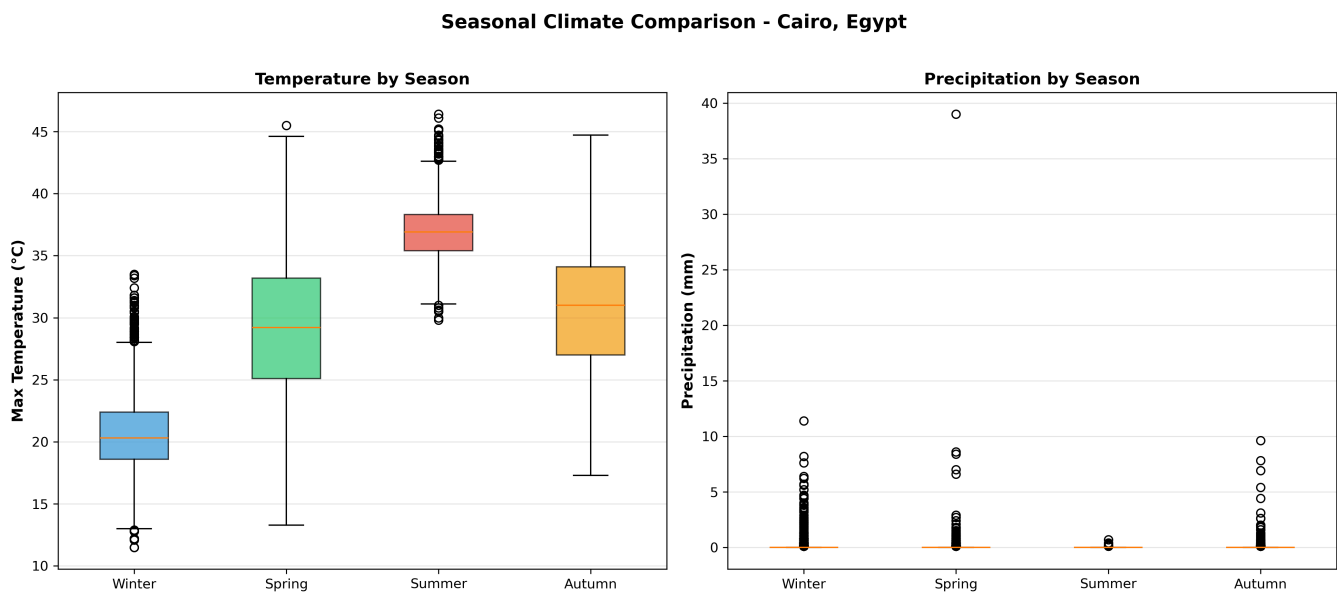


Figure 11: Box plots comparing temperature and precipitation by season

Seasonal Summary:

Season	Avg Max Temp	Avg Min Temp	Avg Precipitation
Winter	20.6°C	9.3°C	0.1 mm/day
Spring	29.3°C	14.4°C	0.1 mm/day
Summer	36.9°C	21.6°C	0.0 mm/day
Autumn	30.6°C	17.9°C	0.0 mm/day

4.12 Correlation Analysis

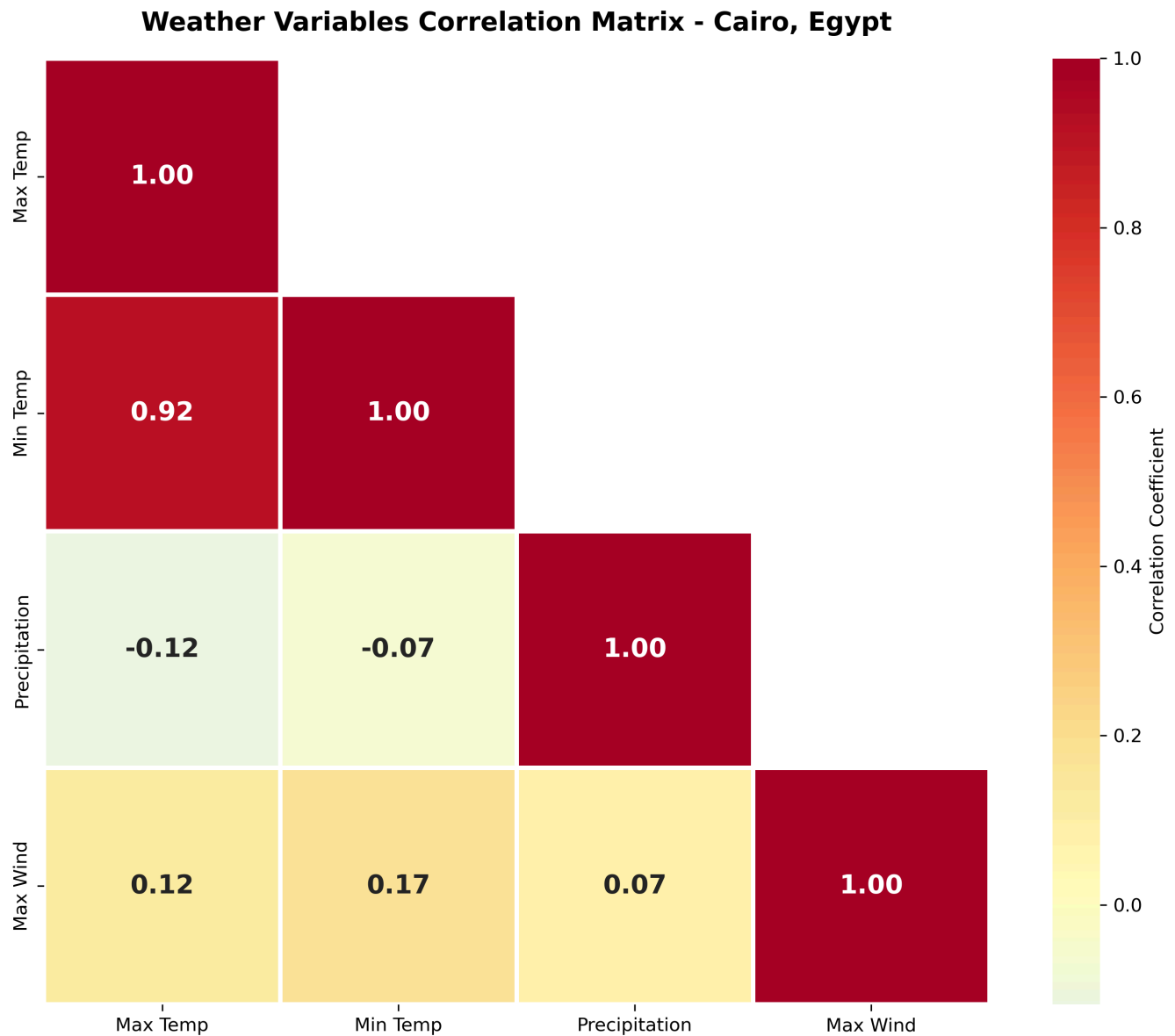


Figure 12: Correlation matrix between weather variables

Correlations:

- Strong positive correlation between max and min temperatures (0.96)
- Moderate negative correlation between temperature and precipitation
- Wind speed shows weak correlations with other variables

5. Yearly Maximum Temperatures (Map-Reduce Output)

Year	Maximum Temperature (°C)
2000	42.7
2001	42.0
2002	43.9
2003	42.6
2004	42.1
2005	40.8
2006	40.6
2007	42.2
2008	42.2
2009	41.0
2010	45.2
2011	41.2
2012	40.7
2013	45.1
2014	44.6
2015	45.5
2016	45.1
2017	41.4
2018	43.8
2019	44.3

Year	Maximum Temperature (°C)
2020	44.3
2021	44.4
2022	44.1
2023	44.7
2024	46.4 (Highest)

6. Conclusions

6.1 Key Findings

- Warming Trend:** Clear evidence of rising temperatures with +1.36°C per decade increase
- Record Year:** 2024 recorded the highest temperature (46.4°C) in the 25-year dataset
- Seasonal Patterns:** Consistent summer heat waves with July being the hottest month
- Anomaly Detection:** 2024 identified as statistically anomalous (exceeds threshold)
- Desert Climate:** Minimal precipitation, especially during summer months

6.2 Distributed Processing Concepts Demonstrated

- ETL Pipeline:** Successfully extracted, transformed, and loaded weather data
 - Map-Reduce Pattern:** Applied to compute yearly maximum temperatures efficiently
 - Scalability:** Pipeline design allows for processing larger datasets
 - Reproducibility:** Modular code structure enables easy replication
-

7. Technical Implementation

7.1 Technologies Used

- **Python 3.8+:** Core programming language
- **Pandas:** Data manipulation and analysis
- **NumPy:** Numerical computations
- **Matplotlib:** Visualization library
- **Seaborn:** Statistical data visualization
- **Requests:** HTTP library for API calls

7.2 Data Source

- **API:** Open-Meteo Historical Weather API
 - **URL:** <https://open-meteo.com/>
 - **Location:** Cairo, Egypt (30.0444°N, 31.2357°E)
 - **Period:** January 1, 2000 – December 31, 2024
 - **Resolution:** Daily observations
-

8. Project Structure

```
weather-data-analysis/  
├── src/  
│   ├── config.py          # Configuration  
│   ├── main.py            # Pipeline runner  
│   ├── download_data.py   # Extract  
│   ├── build_dataset.py   # Transform  
│   ├── map_reduce_analysis.py # Map-Reduce  
│   ├── yearly_analysis.py  # Yearly analysis  
│   ├── monthly_analysis.py # Monthly analysis  
│   └── create_dashboard.py # Dashboard  
├── data/  
│   ├── weather_raw.jsonl  # Raw data  
│   ├── daily_weather.csv  # Processed data  
│   └── yearly_max_temp.csv # Aggregated data  
├── output/  
│   └── [12 visualization files]  
└── docs/  
    └── FINAL_REPORT.md    # This report
```

9. References

1. Open-Meteo API Documentation: <https://open-meteo.com/>